L2 on Lux, or Sovereign L1

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2026

ADOPT-01 • Hanzo AI & Lux Industries • 2026

A five-person team cannot build post-quantum consensus.

Sovereign costs years of cryptography and a validator budget funded from scratch. An L2 inherits both. The init is the same script.

The problem, in Morpheus numbers. Morpheus runs no consensus. The rule that pays a provider executes on a Base diamond Morpheus neither wrote nor secures [verified]. To own that layer, Morpheus goes sovereign, and a sovereign L1 finalizes under a consensus its operator supplies. That consensus must be post-quantum. Lux's is Quasar, a per-round certificate composing a BLS fast-path with lattice (ML-DSA) and hash-based (SLH-DSA) threshold signatures [verified: `luxfi/quasar`]. It took Lux years. Five people ship all native Morpheus code [verified], real revenue is $476K/yr and falling from $3.24M/qtr at the 2024 launch [verified], and there is no cryptography team. A sovereign validator set also sits outside the Lux staking economy, so its security is funded from scratch and the return on MOR bonded for it is forgone.

The fix, named to a repo. `l2-template`. Its `init.sh` stands Morpheus up as an L2 on the Lux primary network in minutes; Hanzo and Zoo were made this way [verified]. Horizon, the finality Quasar produces, applies to Morpheus blocks. Validators are Lux-managed. MOR bonded into the chain joins the shared security economy and earns the Lux staking return. The sovereign path is the same `init.sh` plus one transaction, `ConvertNetworkToL1Tx`; its arithmetic is LP-314. Nothing migrates to begin: the Base diamond stays canonical while the L2 runs beside it.

The outcome, quantified. Post-quantum finality and staking yield are inherited, not rebuilt.

$$\text{Sovereign} = \underbrace{T_{\text{crypto}}}_{\text{multi-year, } \textit{Lux already paid it}} + \underbrace{S \cdot r}_{\text{forgone staking / yr}} \qquad \text{L2} = \text{one } \texttt{init.sh} \text{ run} + \text{MOR bridged for g}$$

T_{crypto} is the Quasar-equivalent, the effort Lux spent over years to reach Horizon [verified: it exists]. S is the MOR bonded for security and r the Lux staking rate: on an L2 the bond earns $S \cdot r$, on a sovereign set it earns nothing [illustrative: S and r are not yet set for Morpheus, so the term is parametric, not a figure]. The L2 side is an afternoon against a floor of one bridge transfer [verified]. The delta the L2 avoids is a multi-year cryptography program plus a forgone yield stream, exchanged for one thing: sovereign control of the validator set.

Sovereign L1

- Own validator set, MOR as native gas, fee policy under its own governance.
- Pays T_{crypto} to build a post-quantum consensus and funds security outside the Lux economy, forgoing $S \cdot r$.

L2 on Lux

- Inherits Horizon finality and Lux-managed validators, no cryptography to build, and earns $S \cdot r$.
- Gives up sovereign control of the validator set to Lux.

The assumption, and why Lux. The sovereign trade pays only if controlling the validator set outweighs $T_{\text{crypto}} + S \cdot r$. For a five-person team whose scarce resource is cryptographic engineering, it does not. Lux is the counterparty because it already produces Horizon and runs the staking economy the bond earns from: the same `init.sh` that made Hanzo and Zoo makes Morpheus, no new cryptography and no foreign bridge. The choice is Morpheus governance's, and nothing else in the series waits on it.

ADOPT-01, the anchor decision for the series. Sovereign-L1 arithmetic in LP-314. • `research@lux.network`